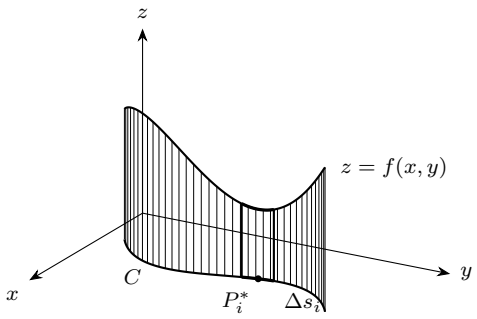
Integrating Along a Curve

The integral $\int_a^b f(x)\,dx$ adds up values of f along an interval, a line segment on the x-axis. Now we integrate along *any* curve C, in the plane or in space. A **scalar line integral** integrates a function f along C; a **vector line integral** integrates a vector field $\mathbf{F}$ along C.

Start with a function $f(x,y) \geq 0$ and a smooth curve C in the xy-plane. Picture a sheet, like a curtain, standing on C and reaching up to the surface $z = f(x,y)$. To find the area of the sheet, chop C into n small pieces with lengths $\Delta s_1, \ldots, \Delta s_n$, and choose a point P_i^* in the ith piece. The strip of the sheet above that piece has area about $f(P_i^*)\,\Delta s_i$. Adding up the strips gives a Riemann sum, except that the pieces are pieces of a curve instead of subintervals of the x-axis.



Definition (Scalar Line Integral). Let f be a function whose domain includes the smooth curve C. The **scalar line integral** of f along C is

if this limit exists. For a curve in the plane and a function of two variables, write $f(x,y)$.

If $f \geq 0$ on a curve C in the plane, $\int_C f\,ds$ is the area of the sheet. Two other meanings come up often. If a thin wire in the shape of C has linear density $\rho(x,y,z)$ (mass per unit length), then its **mass** is $\int_C \rho\,ds$. And with $f = 1$, the sum just adds up the lengths of the pieces, so $\int_C 1\,ds$ is the **arc length** of C.

Evaluating a Scalar Line Integral

To compute, we use a parameterization $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$, $a \leq t \leq b$, that traces C exactly once. From our work with arc length, a short piece of the curve traced over a time Δt has length $\Delta s \approx \|\mathbf{r}'(t)\|\ \Delta t$. So $ds = \|\mathbf{r}'(t)\|\ dt$.

Theorem (Scalar Line Integral Calculation). If f is continuous and $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$, $a \leq t \leq b$, is a smooth parameterization of C that traces it exactly once, then

The value does not depend on which parameterization we use, or on which direction we travel, as long as C is traced exactly once. Most curves in this section are built from a few standard pieces.

Parameterizations to Know.

Circle of radius a centered at the origin, counterclockwise (for an arc, shorten the t-interval):

Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, counterclockwise:

Line segment from (x_0, y_0, z_0) to (x_1, y_1, z_1):

Graph $y = g(x)$ from $x = a$ to $x = b$:

Example 1. Evaluate $\displaystyle\int_C (2 + x^2 y)\, ds$, where C is the upper half of the unit circle $x^2 + y^2 = 1$.

Example 2. Evaluate $\displaystyle\int_C xz\, ds$, where C is the line segment from $(0, 1, 2)$ to $(2, 3, 3)$.

Practice 1.

(a) Evaluate $\displaystyle\int_C x\, ds$, where C is the part of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

(b) A thin wire has the shape of the helix $\mathbf{r}(t) = \langle 3\cos t,\, 3\sin t,\, 4t\rangle$, $0 \le t \le \pi$, and its linear density is $\rho(x, y, z) = z$. Find the mass of the wire.

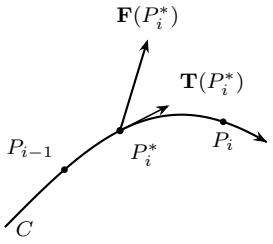
Vector Line Integrals and Work

When a constant force $\mathbf{F}$ moves an object along a straight displacement $\mathbf{D}$, the work done is $W = \mathbf{F} \cdot \mathbf{D}$. Now let $\mathbf{F}$ be a force *field* and let a particle move along a curve C. Two things change: the force can be different at every point, and the path can bend.

The work also depends on the direction of travel. When you walk up a hill, gravity does negative work on you; when you walk down the same path, it does positive work. So we choose a direction along C, called an **orientation**. A curve with an orientation is an **oriented curve**. A curve is **closed** if it has a parameterization with $\mathbf{r}(a) = \mathbf{r}(b)$ that traces it exactly once (it ends where it starts).

Chop C into small pieces. On the piece from P_{i-1} to P_i, the particle moves a distance of about Δs_i in the direction of the unit tangent vector $\mathbf{T}(P_i^*)$. So the displacement is about $\Delta s_i\, \mathbf{T}(P_i^*)$, and the work done on that piece is about

$$\mathbf{F}(P_i^*) \cdot \big(\Delta s_i\, \mathbf{T}(P_i^*)\big) = \big[\mathbf{F}(P_i^*) \cdot \mathbf{T}(P_i^*)\big]\Delta s_i.$$



Definition (Vector Line Integral and Work). The **vector line integral** of the vector field $\mathbf{F}$ along the oriented smooth curve C is

if this limit exists. If $\mathbf{F}$ is a force field, the **work** done by $\mathbf{F}$ in moving a particle along C is

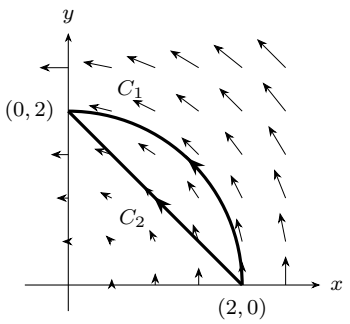
The dot product $\mathbf{F} \cdot \mathbf{T}$ is the component of $\mathbf{F}$ in the direction of motion. It is positive where $\mathbf{F}$ pushes the particle forward, negative where $\mathbf{F}$ pushes it backward, and zero where $\mathbf{F}$ is perpendicular to the path. To compute, use $\mathbf{T} = \mathbf{r}'(t)/\,\|\mathbf{r}'(t)\|$ and $ds = \|\mathbf{r}'(t)\|\, dt$. The two lengths cancel, leaving $\mathbf{F} \cdot \mathbf{T}\, ds = \mathbf{F}\big(\mathbf{r}(t)\big) \cdot \mathbf{r}'(t)\, dt$. For this reason we write $d\mathbf{r} = \mathbf{r}'(t)\, dt$.

Evaluating a Vector Line Integral. If $\mathbf{r}(t)$, $a \le t \le b$, traces C exactly once in the direction of its orientation, then

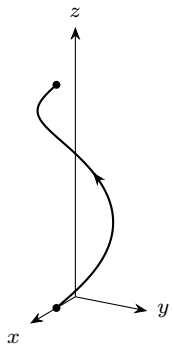
Example 3. Let $\mathbf{F}(x, y) = \langle -y,\, x\rangle$, the rotational field from the vector fields notes. Find the work done by $\mathbf{F}$ on a particle that moves from $(2, 0)$ to $(0, 2)$

(a) along the quarter circle C_1 centered at the origin;

(b) along the line segment C_2.



Example 4. Find the work done by the force field $\mathbf{F}(x, y, z) = \langle -y,\, x,\, z \rangle$ on a particle that moves along the helix $\mathbf{r}(t) = \langle \cos t,\, \sin t,\, t \rangle$, $0 \le t \le 2\pi$.



Practice 2.

(a) Find the work done by the radial field $\mathbf{G}(x, y) = \langle x,\, y \rangle$ along each of the two paths C_1 and C_2 in Example 3. Explain the answer for C_1 using a picture of $\mathbf{G}$.

(b) Evaluate $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y, z) = \langle y,\, z,\, x \rangle$ and C is given by $\mathbf{r}(t) = \langle t,\, t^2,\, t^3 \rangle$, $0 \le t \le 1$.

Orientation and Another Notation

If C is an oriented curve, $-C$ denotes the same curve with the opposite orientation. Traveling the other way flips every unit tangent vector $\mathbf{T}$ to $-\mathbf{T}$, so the vector line integral changes sign. A scalar line integral does not change, because ds measures length, which is never negative:

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_{C} \mathbf{F} \cdot d\mathbf{r}, \qquad \int_{-C} f\, ds = \int_{C} f\, ds.$$

For instance, moving from $(0, 2)$ to $(2, 0)$ along the quarter circle in Example 3, the field $\langle -y, x\rangle$ does work -2π.

There is a second common way to write a vector line integral. If $\mathbf{F} = \langle P, Q, R\rangle$ and we think of $d\mathbf{r}$ as $\langle dx, dy, dz\rangle = \langle x'(t), y'(t), z'(t)\rangle\, dt$, then $\mathbf{F} \cdot d\mathbf{r} = P\, dx + Q\, dy + R\, dz$.

Line Integrals in Differential Form. If $\mathbf{F} = \langle P, Q, R\rangle$ and $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$, $a \le t \le b$, traces C in the direction of its orientation, then

In the plane, drop the $R\, dz$ term. To evaluate, write x, y, z, dx, dy, and dz in terms of t.

Example 5. Evaluate $\displaystyle\int_C xy\, dx + (x - y)\, dy$, where C is the part of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.

Piecewise Smooth Curves

Many curves have corners, like the boundary of a triangle. A curve C is **piecewise smooth** if it is made of finitely many smooth curves $C_1, C_2, \ldots, C_n$, each one starting where the one before it ends. We write $C = C_1 + C_2 + \cdots + C_n$ and integrate over each piece with its own parameterization. The rules below match the ones for $\int_a^b f(x)\, dx$, and property (iv) holds for scalar line integrals $\int_C f\, ds$ too.

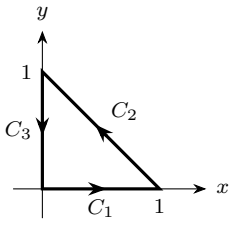
Theorem (Properties of Vector Line Integrals). Let $\mathbf{F}$ and $\mathbf{G}$ be continuous vector fields on the oriented curve C, and let k be a constant.

(i) (ii)

(iii)

(iv) If $C = C_1 + C_2 + \cdots + C_n$ is piecewise smooth, then

Example 6. Evaluate $\displaystyle\int_C x^2\,dx + xy\,dy$, where C is the triangle with vertices $(0,0)$, $(1,0)$, and $(0,1)$, oriented counterclockwise.



Practice 3.

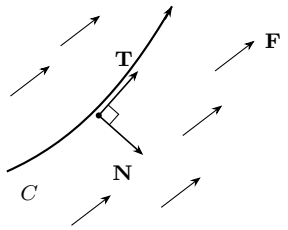
(a) Evaluate $\displaystyle\int_C 2x\,ds$, where C is the parabola $y = x^2$ from $(0,0)$ to $(1,1)$ followed by the vertical line segment from $(1,1)$ to $(1,2)$.

(b) Evaluate $\displaystyle\int_C y\,dx + z\,dy + x\,dz$, where C is the line segment from $(2,0,0)$ to $(3,4,5)$.

Flux and Circulation

Let $\mathbf{F}$ be the velocity field of a fluid flowing in the plane, and let C be a curve in the plane. Think of C as a membrane that the fluid passes through freely. The **flux** of $\mathbf{F}$ across C measures the rate at which fluid crosses C. To measure it, we replace the unit tangent vector $\mathbf{T}$ in $\int_C \mathbf{F} \cdot \mathbf{T}\, ds$ with a unit *normal* vector $\mathbf{N}$.

If $\mathbf{r}(t) = \langle x(t), y(t) \rangle$, then $\mathbf{n}(t) = \langle y'(t),\ -x'(t) \rangle$ is perpendicular to $\mathbf{r}'(t) = \langle x'(t), y'(t) \rangle$ (check the dot product) and points to the *right* as we travel along C. On a closed curve traced counterclockwise, $\mathbf{n}(t)$ points outward.



Definition (Flux). The **flux** of $\mathbf{F}$ across the plane curve C is

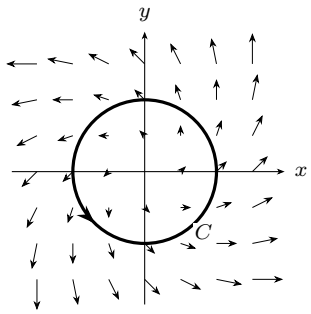
Since $\|\mathbf{n}(t)\| = \|\mathbf{r}'(t)\|$, the lengths cancel again:

Positive flux means the net flow crosses C from left to right (outward, for a closed curve traced counterclockwise). Flux is about flow *across* C. The vector line integral measures flow *along* C, and on a closed curve it gets its own name.

Definition (Circulation). If C is a closed curve, the **circulation** of $\mathbf{F}$ along C is the vector line integral

The circle on the integral sign means C is closed. Circulation measures the tendency of the fluid to move around C in the direction of its orientation.

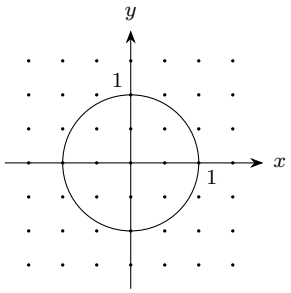
Example 7. Let $\mathbf{F}(x, y) = \langle x - y,\ x + y \rangle$ and let C be the unit circle, oriented counterclockwise. Find the flux of $\mathbf{F}$ across C and the circulation of $\mathbf{F}$ along C.



Practice 4.

(a) Find the flux of $\mathbf{F}(x, y) = \langle y,\, x\rangle$ across the line segment from $(0, 0)$ to $(1, 2)$, oriented from $(0, 0)$ to $(1, 2)$.

(b) Let $\mathbf{F}(x, y) = \langle y,\, 0\rangle$ and let C be the unit circle, oriented counterclockwise. Sketch $\mathbf{F}$ on the grid at the right. Find the flux of $\mathbf{F}$ across C and the circulation of $\mathbf{F}$ along C. Do the signs agree with your sketch?



One last comparison points ahead to the next section.

Example 8. Let $f(x, y) = x^2 y$ and $\mathbf{F} = \nabla f$. Evaluate $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 2)$ along (a) the line segment and (b) the parabola $y = 2x^2$. Compare both answers with $f(1, 2) - f(0, 0)$.

Answers

Practice 1. (a) $\displaystyle\int_0^1 t\sqrt{1 + 4t^2}\, dt = \frac{5\sqrt{5} - 1}{12}$. (b) $\displaystyle\int_0^{\pi} 4t \cdot 5\, dt = 10\pi^2$.

Practice 2. (a) 0 along C_1 and 0 along C_2. On C_1, $\mathbf{G}$ points straight away from the origin, perpendicular to the circle, so $\mathbf{G} \cdot \mathbf{T} = 0$. (b) $\dfrac{89}{60}$.

Practice 3. (a) $\dfrac{5\sqrt{5} - 1}{6} + 2 = \dfrac{5\sqrt{5} + 11}{6}$. (b) $\dfrac{49}{2}$.

Practice 4. (a) $\dfrac{3}{2}$. (b) Flux 0; circulation $-\pi$. The flow is to the right above the x-axis and to the left below it, so it turns clockwise around C.